

Dr. Babasaheb Ambedkar Technological University

Supplementary Summer Examination – 2025

Course: B. Tech.

Branch: Civil Engineering

Semester: VI

Subject Code & Name: **BTCVC602 Y22 Foundation Engineering**

Max Marks: 60

Date: 02/07/2025

Duration: 3 Hrs.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question **No. 1 will be compulsory and include objective-type questions.**
3. Candidates are required to **attempt any four questions from Q. No. 2 to Q. No. 6.**
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

Q. 1	Objective type questions. (Compulsory Question)	(Level /CO)	Marks
1	Undisturbed soil samples are required for conducting ----- a) Consolidation b) Shrinkage limit c) liquid limit d) hydrometer test	CO1	01
2	Undisturbed samples are obtained by ----- a) Thin-walled Samplers b) Thick-walled Samplers c) Augers d) None	CO1	01
3	If two footings are connected by beam it is known as ----- a) Combined Footing b) Mat Footing c) Strap Footing d) Cantilever Footing	CO3	01
4	The weight of hammer used in standard penetration test is ----- a) 50 kg b) 75 kg c) 60 kg d) 65 kg	CO1	01
5	Bulging of soil around the footing will be maximum for ----- a) General Shear b) Local Shear c) Plastic Shear d) Punching Shear	CO2	01
6	If the weight of excavated soil is equal to superimposed load, then foundation is known as ----- a) Balanced b) Floating c) Well d) Raft	CO2	01
7	Minium c/c spacing in friction piles of dia. 'D' as per IS code is ----- a) 2D b) 2.5D c) 3D d) 1.5D	CO4	01
8	The settlement of a group of friction piles as compared to that of a single pile is ---- a) Same b) Less c) More d) None	CO4	01
9	Which of the below is most active clay mineral? a) Na-illite b) Na-kaolinite c) Na-montmorillonite d) Ca-montmorillonite	CO3	01
10	Factor of safety for infinite slope in cohesionless soil _____ with depth a) increase b) decreases c) remains same d) None	CO5	01
11	Depth factor D for base failure of slope is ----- a) Zero b) One c) $0 < D < 1$ d) $D > 1$	CO5	01
12	Radius of friction circle by friction circle method is----- a) $R \cos \phi$ b) $R \sin \phi$ c) $R \tan \phi$ d) $R \sec \phi$	CO5	01

Q. 2 Solve the following.		12
A) Draw the neat labelled sketch of sampler & explain the below terms: 1) Area Ratio, 2) Inside Clearance 3) Outside clearance of the sampler.	CO1	06
B) Explain the Standard Penetration Test?	CO1	06
Q.3 Solve the following.		12
A) State and explain the below terms 1) Gross Bearing capacity, 2) Net Ultimate bearing capacity, and 3) Net Safe bearing capacity	CO2	06
B) Explain the terms in Terzaghi's bearing capacity eq. developed using general shear failure?	CO2	06
Q. 4 Solve <u>Any Two</u> of the following.		12
A) State and explain the following terms: A) Swelling Potential B) Swelling Pressure C) Free Swell	CO3	06
B) Explain the characteristics of expansive soils?	CO3	06
C) Enlist and explain the type's settlement of foundations?	CO3	06
Q.5 Solve <u>Any Two</u> of the following.	CO4	12
A) A group of 9 piles with 3 piles in a row was driven into soft clay extending from ground level to a great depth. The diameter and length of piles were 30 cm and 10 m respectively. The unconfined compressive strength of clay is 70 kN/m^2 . If the piles were spaced at 90 cm c/c , compute the allowable load on the pile group on the basis of shear failure criteria for a factor of safety of 2.5. Neglect bearing at the tip of piles. Take $m = 0.6$ for shear mobilization around each pile.	CO4	06
B) Draw neat sketches of common types of shapes of well foundations.	CO4	06
C) Explain the concept of <i>sheet pile walls</i> in detail.	CO4	06
Q. 6 Solve <u>Any Two</u> of the following.		12
A) What are the different types of slopes. Explain in detail with neat sketch and example?	CO5	06
B) Explain any one of the method to analyze the stability of slopes.	CO5	06
C) A slope is to be constructed at an inclination of 30° with the horizontal. Determine the safe height of the slope at factor of safety of 2.0. The soil has the following properties: $c = 25 \text{ kN/m}^2$, $\phi = 22.5^\circ$ and $g = 20 \text{ kN/m}^3$	CO5	06

*** End ***